

Differential Equations

imp for finals

Ex - 2.8

CLO - 1

2.10

1.9

$y_p =$ particular solution for
non homogeneous eq

$y_h =$ general solution for
homogeneous eq

Ex-2.8

Q: 1, 2, 6, 7, 9, 10

Q: $y'' - y = 3e^{2x}$

$y_p = e^{2x}$

Solⁿ

$y_p = e^{2x}$

$y_p' = 2e^{2x}$

$y_p'' = 4e^{2x}$

$4e^{2x} - e^{2x} = 3e^{2x}$

$3e^{2x} = 3e^{2x}$

Yes the given eq is a solution of the given differential eq.

$y = y_H + y_p$ — (A)

for y_H take L.H.S = 0

$y_H = y'' - y = 0$

let $y'' = m^2$, $y' = m$ & $y = 1$

$m^2 - 1 = 0$

$m^2 = 1$

$m = \pm 1$

roots are different

$y_H = c_1 e^x + c_2 e^{-x}$

put in (A)

$y = c_1 e^x + c_2 e^{-x} + e^{2x}$

$$② \quad y'' - y' - 2y = -4x \quad y_p = 2x - 1$$

Soln

$$y_p = 2x - 1$$

$$y_p' = 2$$

$$y_p'' = 0$$

$$0 - 2 - 2(2x - 1) = -4x$$

$$-2 - 4x + 2 = -4x$$

$$-4x = -4x$$

Yes the given eq is a solution of the differential eq

$$y = y_h + y_p$$

$$y_h = y'' - y' - 2y = 0$$

$$\text{let } y'' = m^2 \quad y' = m \quad y = 1$$

$$m^2 - m - 2 = 0$$

$$m^2 - 2m + m - 2 = 0$$

$$m(m-2) + 1(m-2) = 0$$

$$(m+1)(m-2) = 0$$

$$m = -1$$

$$m = 2$$

$$y_h = c_1 e^{-x} + c_2 e^{2x}$$

$$y = c_1 e^{-x} + c_2 e^{2x} + 2x - 1$$

$$(6) (D^2 + 3D - 4)y = -6.8 \sin x$$

$$y_p = \sin x + 0.6 \cos x$$

Soln

$$y_p = \sin x + 0.6 \cos x$$

$$y_p' = \cos x - 0.6 \sin x$$

$$y_p'' = -\sin x - 0.6 \cos x$$

$$y'' + 3y' - 4y = -6.8 \sin x$$

$$-\sin x - 0.6 \cos x + 3(\cos x - 0.6 \sin x) - 4(\sin x + 0.6 \cos x) = -6.8 \sin x$$

$$-\sin x - 0.6 \cos x + 3 \cos x - 1.8 \sin x - 4 \sin x - 2.4 \cos x = -6.8 \sin x$$

$$\sin x (-1 - 1.8 - 4) + \cos x (-0.6 + 3 - 2.4) = -6.8 \sin x$$

$$\begin{aligned} & \quad \quad \quad (0) \\ -6.8 \sin x & - 0 \cos x = -6.8 \sin x \\ \text{L.H.S} & = R.H.S \end{aligned}$$

Let

$$y'' + 3y' - 4y = 0$$

$$m^2 + 3m - 4 = 0$$

$$m^2 + 4m - 1(m+4) = 0$$

$$m(m+4) - 1(m+4) = 0$$

$$(m+4)(m-1) = 0$$

$$m = -4$$

$$m = 1$$

$$y_h = c_1 e^{-4x} + c_2 e^x$$

$$y = c_1 e^{-4x} + c_2 e^x + \sin x + 0.6 \cos x$$

$$(7) \quad (x^2 D^2 - 2xD + 2)y = 5x^3 \cos x$$

$$y_p = -5x \cos x$$

Soln:

$$y_p = -5x \cos x$$

$$y_p' = -5(-x \sin x + \cos x) \Rightarrow 5x \sin x - 5 \cos x$$

$$y_p'' = -5(-x \cos x + \sin x) - \sin x$$

$$= 5x \cos x + 5 \sin x + 5 \sin x$$

$$= 5x \cos x + 10 \sin x$$

$$x^2 y'' - 2xy' + 2y = 5x^3 \cos x$$

$$5x^3 \cos x + 10x^2 \sin x - 10x^2 \sin x + 10x \cos x$$

$$- 10x \cos x = 5x^3 \cos x$$

$$5x^3 \cos x = 5x^3 \cos x$$

Hence the given eqn is a sol of diff eq

$$\text{let } x^2 y'' - 2xy' + 2y = 0$$

$$m(m-1) - 2m + 2 = 0$$

$$m^2 - m - 2m + 2 = 0$$

$$m^2 - 3m + 2 = 0$$

$$m^2 - 2m - 1m + 2 = 0$$

$$m(m-2) - 1(m-2) = 0$$

$$(m-1)(m-2) = 0$$

$$m = 1$$

$$m = 2$$

$$y_H = C_1 x + C_2 x^2$$

$$y = y_H + y_p$$

$$= C_1 x + C_2 x^2 - 5x \cos x$$

Q. $(4x^2 D^2 + 1) y = (1-x^2) \cos 0.5x$

$$y_p = \cos 0.5x$$

Soln

$$y_p = \cos 0.5x$$

$$y_p' = -\sin(0.5x) (0.5x)'$$

$$y_p' = -0.5 \sin(0.5x)$$

$$y_p'' = -0.5 \cos(0.5x) (0.5x)'$$

$$y_p'' = -0.25 \cos(0.5x)$$

$$4x^2 y'' + y = (1-x^2) \cos(0.5x)$$

~~$$y = \frac{1}{4} (m(m-1))$$~~

$$4x^2 (-0.25 \cos(0.5x)) + \cos 0.5x = (1-x^2) \cos(0.5x)$$

$$-x^2 \cos(0.5x) + \cos(0.5x) = (1-x^2) \cos(0.5x)$$

$$(1-x^2) \cos(0.5x)$$

the given eq is a solution

$$4x^2 y'' + y = 0$$

$$4m(m-1) + 1 = 0$$

$$4m^2 - 4m + 1 = 0$$

using quadratic formula $m = \frac{1}{2}$

$$y_H = (c_1 + c_2 \ln x) x^{1/2}$$

$$y = (c_1 + c_2 \ln x) x^{1/2} + \cos 0.5x$$



Ex - 2.10

1, 2, 3, 4, 5, 6

$$\textcircled{1} \quad y'' - 2y' + y = x^{3/2} e^x$$

Solⁿ compare with

$$y'' + P(x)y' + Q(x)y = R(x)$$

$\Rightarrow$ Auxiliary equation

$$\text{Let } y'' - 2y' + y = 0$$

$$m^2 - 2m + 1 = 0$$

$$m^2 - m - m + 1 = 0$$

$$m(m-1) - 1(m-1) = 0$$

$$(m-1)(m-1) = 0$$

$$m = 1$$

roots are same

$$y_H = (c_1 + c_2 x) e^{1 \cdot x}$$

$$y_H = c_1 e^x + c_2 x e^x$$

$$\Rightarrow \text{let } e^x = y_1 \quad \text{and } x e^x = y_2$$

$$\text{now } y_1' = e^x \quad \text{and } y_2' = x e^x + e^x$$

$$\Rightarrow \text{Wronskian Formula, } W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1'$$

$$\therefore W = \begin{vmatrix} e^x & x e^x \\ e^x & x e^x + e^x \end{vmatrix} = e^x (x e^x + e^x) - e^x (x e^x)$$

$$W = x e^{2x} + e^{2x} - x e^{2x}$$

$$W = e^{2x}$$

$\Rightarrow$

$$y_p = -y_1 \int \frac{y_2 R(x)}{W} dx + y_2 \int \frac{y_1 R(x)}{W} dx$$

put values

$$\begin{aligned} y_p &= -e^x \int \frac{x e^x (x^{3/2} e^x)}{e^{2x}} dx + x e^x \int \frac{e^x x^{3/2} e^x}{e^{2x}} dx \\ &= -e^x \int \frac{x^{5/2} e^{2x}}{e^{2x}} dx + x e^x \int \frac{x^{3/2} e^{2x}}{e^{2x}} dx \\ &= -e^x \int x^{5/2} dx + x e^x \int x^{3/2} dx \end{aligned}$$

$$= -e^x \frac{x^{7/2}}{7/2} + x e^x \frac{x^{5/2}}{5/2} + C$$

$$y_p = -\frac{2e^x x^{7/2}}{7} + \frac{2x e^x x^{5/2}}{5} + C$$

as $y = y_h + y_p$

$$\therefore y = (c_1 + c_2 x) e^x - \frac{2}{7} e^x x^{7/2} + \frac{2}{5} x e^x x^{5/2} + C$$

$$y =$$

$$\textcircled{2} \quad y'' + 4y = 2\sec 2x$$

Solⁿ compare with

$$y'' + P(x)y' + a(x)y = r(x)$$

auxiliary eq

$$y'' + 4y = 0$$

$$m^2 + 4 = 0$$

$$m^2 = -4$$

$$m = \cancel{2} \pm 2i$$

roots are ~~different~~ complex

let ~~$y = c_1 e^{-2x} + c_2 e^{2x}$~~
 ~~$y_1 = e^{-2x}$ and $y_2 = e^{2x}$~~
 ~~$y_1' = -2e^{-2x}$ $y_2' = 2e^{2x}$~~

Wronskian formula $w = \begin{vmatrix} e^{-2x} & e^{2x} \\ -2e^{-2x} & 2e^{2x} \end{vmatrix} = e^{-2x}(2e^{2x}) - e^{2x}(-2e^{-2x})$
 ~~$w = 2e^{-2x-2x}$~~

$$y_H = c_1 \cos 2x + c_2 \sin 2x$$

$$y_1 = \cos 2x$$

$$y_2 = \sin 2x$$

$$y_1' = -2\sin 2x$$

$$y_2' = 2\cos 2x$$

Wronskian formula $\Rightarrow$

$$w = \begin{vmatrix} \cos 2x & \sin 2x \\ -2\sin 2x & 2\cos 2x \end{vmatrix}$$

$$w = \cos 2x (2\cos 2x) - \sin 2x (-2\sin 2x)$$

$$w = 2\cos^2 2x + 2\sin^2 2x$$

$$w = 2(\cos^2 2x + \sin^2 2x)$$

$$w = 2$$

$$\begin{aligned}
 y_p &= -y_1 \int \frac{y_2 r(x)}{w} dx + y_2 \int \frac{y_1 r(x)}{w} dx \\
 &= -\cos 2x \int \frac{\sin 2x \cdot 2 \sec 2x}{2} dx + \sin 2x \int \frac{\cos 2x \cdot 2 \sec 2x}{2} dx \\
 &= \frac{\cos 2x}{2} \int -2 \sin 2x \times \frac{1}{\cos 2x} dx + \sin 2x \int \cos 2x \times \frac{1}{\cos 2x} dx \\
 y_p &= \frac{\cos 2x}{2} \ln |\cos 2x| + x \sin 2x + C
 \end{aligned}$$

$$y = y_H + y_p$$

$$y = C_1 \cos 2x + C_2 \sin 2x + \frac{1}{2} \cos 2x \ln |\cos 2x| + x \sin 2x + C$$

$$(3) \quad y'' - 2y' + y = 12e^x / x^3$$

Soln

$$y'' + P(x)y' + Q(x)y = R(x)$$

$$y'' - 2y' + y = 0$$

$$m^2 - 2m + 1 = 0$$

$$m^2 - m - m + 1 = 0$$

$$m(m-1) - 1(m-1) = 0$$

$$(m-1)(m-1)$$

$$m=1$$

roots are same

$$y_h = (c_1 + c_2 x) e^x$$

$$y_h = c_1 e^x + c_2 x e^x$$

let

$$y_1 = e^x$$

and

$$y_2 = x e^x$$

$$y_1' = e^x$$

$$y_2' = x e^x + e^x$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^x & x e^x \\ e^x & x e^x + e^x \end{vmatrix}$$

$$W = e^x (x e^x + e^x) - e^x (x e^x)$$

$$W = x e^{2x} + e^{2x} - x e^{2x}$$

$$W = e^{2x}$$

$$y_p = -e^x \int \frac{x e^x \cdot 12e^x / x^3}{e^{2x}} dx + x e^x \int \frac{e^x \cdot 12e^x}{x^3 e^{2x}} dx$$

$$y_p = -e^x \int \frac{12x e^{2x}}{x^3 e^{2x}} dx + x e^x \int \frac{12 e^{2x}}{x^3 e^{2x}} dx$$

$$y_p = -e^x \int \frac{12}{x^2} dx + x e^x \int \frac{12}{x^3} dx$$

$$y_p = -12e^x \int x^{-2} dx + 12x e^x \int x^{-3} dx$$

$$y_p = -12e^x \frac{x^{-1}}{-1} + 12x e^x \frac{x^{-2}}{-2} + C$$

$$y_p = 12 \frac{e^x}{x} - 6 \frac{e^x}{x} + C$$

$$y_p = 6 \frac{e^x}{x} + C$$

$$y = y_h + y_p$$

$$y = c_1 e^x + c_2 x e^x + 6 \frac{e^x}{x} + C$$

$$\textcircled{1} \quad y'' + y = \csc x$$

Solⁿ: compare with $y'' + p(x)y' + q(x)y = r(x)$

let $y'' + y = 0$

$$m^2 + 1 = 0$$

$$m^2 = -1$$

$$m = \pm i$$

$$m(m+1) = 0$$

$$m = 0 \quad \text{or} \quad m+1 = 0$$

$$y_h = c_1 e^{0x} + c_2 e^{-x}$$

$$y_h = c_1 + c_2 e^{-x}$$

$$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix}$$

$$W = \cos^2 x + \sin^2 x$$

$$W = 1$$

$$y_p = -\cos x \int \frac{\sin x \cdot \csc x}{1} dx + \sin x \int \frac{\cos x \cdot \csc x}{1} dx$$

$$y_p = -\cos x \int \frac{\sin x}{\sin x} dx + \sin x \int \frac{\cos x}{\sin x} dx$$

$$y_p = -\cos x \int dx + \sin x \ln(\sin x) + C$$

$$y_p = -\cos x(x) + \sin x \ln(\sin x) + C$$

$$y = y_h + y_p$$

$$y = c_1 \cos x + c_2 \sin x - x \cos x + \sin x \ln(\sin x) + C$$

$$y_h = c_1 \cos x + c_2 \sin x$$

$$y_1 = \cos x$$

$$y_1' = -\sin x$$

$$y_2 = \sin x$$

$$y_2' = \cos x$$

Chp-11

Q1: $ux + uy = 0$

sol let $u(x, y) = f(x) \times g(y)$ — ^{keypoint} ①

diff w.r.t x

$$ux = f'(x) g(y)$$

diff w.r.t y

$$uy = f(x) g'(y)$$

put in que

$$f'(x) g(y) + f(x) g'(y) = 0$$

$$f'(x) g(y) = -f(x) g'(y)$$

$$\frac{f'(x)}{f(x)} = - \frac{g'(y)}{g(y)} = k$$

$$\frac{f'(x)}{f(x)} = k$$

integrate

$$\int \frac{f'(x)}{f(x)} dx = \int k dx$$

$$\ln(f(x)) = kx + c_1$$

Taking e on b.s

$$e^{\ln(f(x))} = e^{kx + c_1}$$

$$f(x) = e^{kx + c_1}$$

put in key point

$$\frac{g'(y)}{g(y)} = -k$$

integrate

$$\int \frac{g'(y)}{g(y)} dy = - \int k dy$$

$$\ln(g(y)) = -ky + c_2$$

taking e on b.s

$$e^{\ln(g(y))} = e^{-ky + c_2}$$

$$g(y) = e^{-ky + c_2}$$

$$\ln a + \ln b = \ln ab$$

$$a \ln b = \ln b^a$$

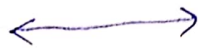
$$u(x, y) = f(x) g(y)$$

$$= e^{kx+c_1} e^{-ky+c_2}$$

$$= e^{kx+c_1 - ky+c_2}$$

$$= e^{k(x-y) + c}$$

$$= C e^{k(x-y)}$$



Q2: $x u_x + y u_y = 0$

let $u(x, y) = f(x) g(y)$ — (1) keypoint

diff w.r.t (x)

$$u_x = f'(x) g(y)$$

diff w.r.t (y)

$$u_y = f(x) g'(y)$$

$$x f'(x) g(y) - y f(x) g'(y) = 0$$

$$x f'(x) g(y) = y f(x) g'(y)$$

$$x \frac{f'(x)}{f(x)} = y \frac{g'(y)}{g(y)} = k$$

$$\frac{f'(x)}{f(x)} = \frac{k}{x}$$

$$\frac{g'(y)}{g(y)} = \frac{k}{y}$$

differentiate integrate

$$\ln(f(x)) = k \ln x + \ln c_1$$

$$\ln(f(x)) = \ln x^k + \ln c_1$$

$$\ln(f(x)) = \ln x^k c_1$$

$$f(x) = x^k c_1$$

$$\ln(g(y)) = k \ln y + \ln c_2$$

$$\ln(g(y)) = \ln y^k + \ln c_2$$

$$\ln(g(y)) = \ln y^k c_2$$

$$g(y) = y^k c_2$$

$$f(x) g(y) = x^k c_1 y^k c_2$$

$$= (xy)^k C$$

$$\therefore c_1 c_2 = C$$

Q3: $ux + uy = 2(x+y)u$

Sol: let $u(x,y) = f(x) g(y)$ — (1)
diff w.r.t 'x'

$$ux = f'(x) g(y)$$

diff wrt 'y'

$$uy = f(x) g'(y)$$

put value

$$f'(x) g(y) + f(x) g'(y) = 2(x+y) f(x) g(y)$$

$$\frac{f'(x) g(y)}{f(x) g(y)} + \frac{f(x) g'(y)}{f(x) g(y)} = \frac{2(x+y) f(x) g(y)}{f(x) g(y)}$$

$$\frac{f'(x)}{f(x)} + \frac{g'(y)}{g(y)} = 2x + 2y$$

$$\frac{f'(x)}{f(x)} - 2x = -\frac{g'(y)}{g(y)} + 2y = k$$

$$\frac{f'(x)}{f(x)} - 2x = k$$

$$\frac{g'(y)}{g(y)} - 2y = -k$$

integrate

$$\ln(f(x)) - \frac{2x^2}{2} = kx + c_1$$

$$\ln(g(y)) - \frac{2y^2}{2} = -ky + c_2$$

$$\ln(f(x)) - x^2 = kx + c_1$$

$$\ln(g(y)) - y^2 = -ky + c_2$$

$$\ln(f(x)) = x^2 + kx + c_1$$

$$\ln(g(y)) = y^2 - ky + c_2$$

$$f(x) = e^{x^2 + kx + c_1}$$

$$g(y) = e^{y^2 - ky + c_2}$$

$$f(x) g(y) = e^{x^2 + kx + c_1} e^{y^2 - ky + c_2}$$

$$f(x) g(y) = e^{x^2 + y^2 + k(x-y) + c}$$

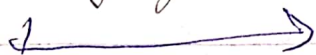
Q4: $u_{xy} - u = 0$

Let $u(x, y) = f(x) g(y)$

$u_x = f'(x) g(y)$

$u_y = f(x) g'(y)$

$y u_{xy} = f(x) g'(y)$



$u = (1 - 2xy + y^2)^{-1/2}$

Prove

$x \frac{\partial u}{\partial x} - y \frac{\partial u}{\partial y} = y^2 u^3$

Given

$u = (1 - 2xy + y^2)^{-1/2}$

Diff. w.r.t. x

$\frac{\partial u}{\partial x} = -\frac{1}{2} (1 - 2xy + y^2)^{-\frac{1}{2}-1} \frac{d}{dx} (1 - 2xy + y^2)$

$= -\frac{1}{2} (1 - 2xy + y^2)^{-\frac{3}{2}} (0 - 2y + 0)$

$= +\frac{1}{2} \times 2y (1 - 2xy + y^2)^{-3/2}$

$\frac{\partial u}{\partial x} = y (1 - 2xy + y^2)^{-3/2}$

$x \frac{\partial u}{\partial x} = xy (1 - 2xy + y^2)^{-3/2}$

$$u = (1 - 2xy + y^2)^{-1/2}$$

Diff w.r.t. y

$$\frac{\partial u}{\partial y} = -\frac{1}{2} (1 - 2xy + y^2)^{-1/2 - 1} \frac{d}{dy} (1 - 2xy + y^2)$$

$$= -\frac{1}{2} (1 - 2xy + y^2)^{-3/2} (0 - 2x + 2y)$$

$$= +\frac{2(x - y)}{2} (1 - 2xy + y^2)^{-3/2}$$

$$\frac{\partial u}{\partial y} = (x - y) (1 - 2xy + y^2)^{-3/2}$$

$$y \frac{\partial u}{\partial y} = xy - y^2 (1 - 2xy + y^2)^{-3/2}$$

$$x \frac{\partial u}{\partial x} - y \frac{\partial u}{\partial y} =$$

$$xy (1 - 2xy + y^2)^{-3/2} - xy + y^2 (1 - 2xy + y^2)^{-3/2}$$

$$= xy(1-2xy+y^2)^{-3/2} - y(x+y)(1-2xy+y^2)^{-3/2}$$

$$= (1-2xy+y^2)^{-3/2} (xy - xy - y^2)$$

$$= y^2(1-2xy+y^2)^{-3/2}$$

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = y^2 u^{-3/2} = y^2 (u^{3/2})^3$$

$$= y^2 u^3 \text{ proved}$$

$$\frac{2-1-2u}{}$$

δ

δ

Ans

①:

prove that

$$u = (1 - 2xy + y^2)^{-1/2}$$

$$x \frac{\partial u}{\partial x} - y \frac{\partial u}{\partial y} = y^2 u^3$$

Ans. $u = (1 - 2xy + y^2)^{-1/2}$

$$\frac{\partial u}{\partial x} = \frac{-1}{2} (1 - 2xy + y^2)^{-3/2} (-2y)$$

$$x \frac{du}{dx} = xy (1 - 2xy + y^2)^{-3/2}$$

$$u = (1 - 2xy + y^2)^{-1/2}$$

$$\frac{\partial u}{\partial y} = \frac{-1}{2} (1 - 2xy + y^2)^{-3/2} (-2x + 2y)$$

$$\frac{\partial u}{\partial y} = \frac{-2(x-y)}{-2} (1 - 2xy + y^2)^{-3/2}$$

$$y \cdot \frac{\partial u}{\partial y} = y(x-y) (1 - 2xy + y^2)^{-3/2}$$

$$x \cdot \frac{du}{dx} - y \frac{\partial u}{\partial y} = xy (1 - 2xy + y^2)^{-3/2} - (xy - y^2) (1 - 2xy + y^2)^{-3/2}$$

$$= (1 - 2xy + y^2)^{-3/2} (xy - xy + y^2)$$

$$= y^2 (1 - 2xy + y^2)^{-3/2}$$

$$= y^2 \left((1 - 2xy + y^2)^{-1/2} \right)^3 \Rightarrow y^2 u^3$$



Differential Eq imp

2 Q's from CLO-1
1.9
2.8
2.10

2 Q's from CLO-3 pics from 23 Dec

1 imp que from 2-1-24 CLO-3

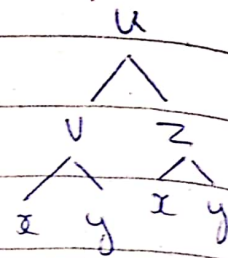
2 Q's from CLO-3 11.4

Ex-11.4

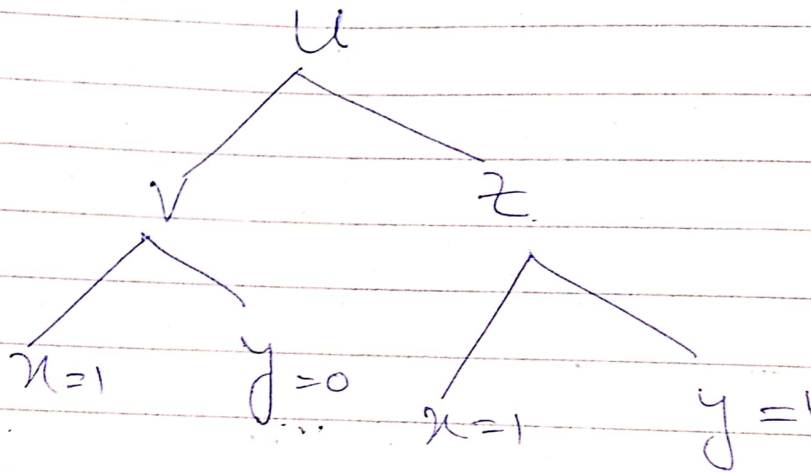
Q: $u_{xy} - u_{yy} = 0$

Q: $v = x \quad z = x + y$

Sol:



$$\begin{aligned} v_x &= 1 & z_x &= 1 \\ v_y &= 0 & z_y &= 1 \end{aligned}$$



$u_{xy} = ?$

$u_x = ?$

$$\begin{aligned} u_x &= \frac{\partial u}{\partial v} \cdot \frac{\partial v}{\partial x} + \frac{\partial u}{\partial z} \cdot \frac{\partial z}{\partial x} \\ &= \frac{\partial u}{\partial v} (1) + \frac{\partial u}{\partial z} (1) \end{aligned}$$

$u_x = u_v + u_z$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial v} \cdot \frac{\partial v}{\partial y} + \frac{\partial u}{\partial z} \cdot \frac{\partial z}{\partial y}$$

$$= \frac{\partial u}{\partial v}(0) + \frac{\partial u}{\partial z} \cdot \frac{\partial z}{\partial y}$$

$$= 0 + u_z(1)$$

$$\boxed{u_y = u_z}$$

$$\boxed{u_{yy} = u_{zz}} \quad \checkmark$$

$$u_{xy} = ?$$

$$(u_x)_y = (u_v + u_z)_y$$

$$u_{vy} + u_{zy}$$

$$= 0 + u_z \cdot \frac{\partial z}{\partial y}$$

$$\boxed{u_{xy} = u_{vz} + u_{zz} \cdot \frac{\partial z}{\partial y}}$$

$$v = x$$

$$z = x + y$$

$$\frac{\partial v}{\partial x} = 1$$

$$\frac{\partial z}{\partial x} = 1$$

$$\frac{\partial u}{\partial z} \quad \frac{\partial z}{\partial y}$$

$$\frac{\partial v}{\partial y} = 0$$

$$\frac{\partial z}{\partial y} = 1$$

Q3

$$u_{xy} - u_{yx} = 0$$

$$v = x$$

$$z = x + y$$

$$\frac{\partial v}{\partial x} = 1$$

$$\frac{\partial z}{\partial x} = 1$$

Soln

$$u_x = \frac{\partial u}{\partial v} \frac{\partial v}{\partial x} + \frac{\partial u}{\partial z} \frac{\partial z}{\partial x}$$

$$u_x = \frac{\partial u}{\partial v} + \frac{\partial u}{\partial z}$$

$$u_x = u_v + u_z$$

$$u_{xy} = (u_v + u_z) \left[\frac{\partial u}{\partial v} \frac{\partial v}{\partial y} + \frac{\partial u}{\partial z} \frac{\partial z}{\partial y} \right]$$

$$u_{xy} = (u_v + u_z) \left[0 + \frac{\partial u}{\partial z} \right]$$

$$u_{xy} = (u_v + u_z) u_z$$

$$u_{xy} = u_{vz} + u_{zz}$$

$$u_y = \frac{\partial u}{\partial v} \frac{\partial v}{\partial y} + \frac{\partial u}{\partial z} \frac{\partial z}{\partial y}$$

$$u_y = 0 + \frac{\partial u}{\partial z}$$

$$u_y = u_z \Rightarrow$$

$$u_{yy} = u_z (v_y + z_y)$$

$$u_{yy} = u_z v_y + u_z z_y \Rightarrow u_{yy} = 0 + u_z$$

$$u_{yy} = u_{zz}$$

$$u_{xx} = u_v + u_z$$

$$u_{xx} = (u_v + u_z) \left[\frac{\partial u}{\partial v} \frac{\partial v}{\partial x} + \frac{\partial u}{\partial z} \frac{\partial z}{\partial x} \right]$$

$$u_{xx} = (u_v + u_z) \left[\frac{\partial u}{\partial v} + \frac{\partial u}{\partial z} \right]$$

$$u_{xx} = (u_v + u_z) \cdot (u_v + u_z)$$

$$u_{xx} = (u_v + u_z)^2 \quad \star$$

$$u_{xx} = u_{vv} + 2u_{vz} + u_{zz}$$

Fourier Series.

$f(x)$ is a function

define $-\pi < x < \pi$

integrable, can be written

$$f(x) = \frac{1}{2} \underline{a_0} + \sum_{n=1}^{\infty} (\underline{a_n} \cos nx + \underline{b_n} \sin nx)$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx$$

Q/ $f(x) = \begin{cases} 0 & -\pi < x < 0 \\ 1 & 0 < x < \pi \end{cases}$

Sol.

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$= \frac{1}{\pi} \left[\int_{-\pi}^0 0 dx + \int_0^{\pi} 1 dx \right]$$

$$= \frac{1}{\pi} \left[0 + [x]_0^{\pi} \right] = \frac{1}{\pi} [\pi - 0]$$

$$\boxed{a_0 = 1}$$

= 1

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx$$

$$= \frac{1}{\pi} \left[\int_{-\pi}^0 0 \cdot \cos nx \, dx + \int_0^{\pi} 1 \cdot \cos nx \, dx \right]$$

$$= \frac{1}{\pi} \left[0 + \int_0^{\pi} \cos nx \, dx \right]$$

$$= \frac{1}{\pi} \left[\frac{\sin nx}{n} \right]_0^{\pi}$$

$$\begin{aligned} \int \cos x \, dx &= \sin x + c \\ \int \cos nx \, dx &= \frac{\sin nx}{n} + c \end{aligned}$$

$$= \frac{1}{\pi n} \left[\sin n\pi - \sin n(0) \right]$$

$$= \frac{1}{\pi n} (0 - 0) = 0$$

$$\boxed{a_0 = 0}$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx$$

$$= \frac{1}{\pi} \left[\int_{-\pi}^0 0 \sin nx \, dx + \int_0^{\pi} 1 \cdot \sin nx \, dx \right]$$

$$\frac{1}{\pi} \left[\left(-\frac{\cos n\pi}{n} \right) \right]_0^{\pi}$$

$$-\frac{1}{n\pi} \left[\cos n\pi - \cos n(0) \right]$$

$$\left(-\frac{1}{n\pi} \left[-1 - (1) \right] \right) = +\frac{2}{n\pi}$$

$$b_n = -\frac{1}{n\pi} \left[(-1)^n - \cos(0) \right]$$

$$n=1$$

$$b_1 = -\frac{1}{\pi} \left[(-1)^1 - \cos(0) \right]$$

$$-\frac{1}{\pi} \left[-1 - 1 \right] = \frac{2}{\pi}$$

$$n=2$$

$$b_2 = -\frac{1}{2\pi} \left[(-1)^2 - \cos(0) \right]$$

$$= -\frac{1}{2\pi} \left[1 - 1 \right] = 0$$

$$b_2 = 0$$

$$\underline{n=3}$$

$$b_3 = \frac{1}{3\pi} [(-1)^3 - \cos(0)]$$

$$b_3 = \frac{1}{3\pi} [-1 - 1] = -\frac{2}{3\pi}$$

$$n=4$$

$$b_4 = 0$$

$$b_5 = -\frac{2}{5\pi}$$

$$b_6 = 0$$

Now

$$f(x) = \frac{1}{2} + 0 + \frac{2}{\pi} \sin x + 0 - \frac{2}{3\pi} \sin 3x$$

$$+ 0 - \frac{2}{5\pi} \sin 5x + \dots$$

$$f(x) = \frac{1}{2} + \frac{2}{\pi} \sin x - \frac{2}{3\pi} \sin 3x - \frac{2}{5\pi} \sin 5x + \dots$$

$$\textcircled{17}: f(x) = \begin{cases} 0, & -\pi \leq x < 0 \\ x, & 0 \leq x \leq \pi \end{cases}$$

Soln

$$f(x) = \frac{1}{2} a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx$$

$$a_0 = \frac{1}{\pi} \left[\int_{-\pi}^0 0 dx + \int_0^{\pi} x dx \right]$$

$$= \frac{1}{\pi} \left[0 + \left[\frac{x^2}{2} \right]_0^{\pi} \right]$$

$$= \frac{1}{\pi} \left[\frac{\pi^2}{2} - \frac{0^2}{2} \right]$$

$$a_0 = \frac{\pi}{2}$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx$$

$$a_n = \frac{1}{\pi} \left[\int_{-\pi}^0 0 \cos(n \cdot 0) dx + \int_0^{\pi} x \cos nx dx \right]$$

$$a_n = \frac{1}{\pi} \left[0 + \left[\frac{x \sin nx}{n} + \frac{\cos nx}{n^2} \right]_0^{\pi} \right]$$

$$a_n = \frac{1}{\pi} \left[\frac{\pi}{n} \sin(n\pi) + \frac{\cos(n\pi)}{n^2} - \left(\frac{\cos(0)}{n^2} \right) \right]$$

$\begin{array}{l} x + \cos nx \\ \sin nx \\ n \\ -\cos nx \\ n^2 \end{array}$

$$a_n = \frac{1}{\pi} \left[\frac{\pi}{n} \sin(n\pi) + \frac{\cos(n\pi)}{n^2} - \frac{1}{n^2} \right]$$

$$a_n = \frac{1}{n} \sin(n\pi) + \frac{\cos(n\pi)}{\pi n^2} - \frac{1}{n^2}$$

$$\underline{n=1} \quad a_1 = 0 + \frac{1}{\pi} - 1 \Rightarrow a_1 = \frac{-1}{\pi} - 1$$

$$\underline{n=2} \quad a_2 = \frac{1}{2} \sin(2\pi) + \frac{\cos(2\pi)}{4\pi} - \frac{1}{4}$$

$$a_2 = 0 + \frac{1}{4\pi} - \frac{1}{4} \Rightarrow a_2 = \frac{1}{4\pi} - \frac{1}{4}$$

$$\underline{n=3} \quad a_3 = 0 + \frac{1}{9\pi} - \frac{1}{9} \Rightarrow a_3 = \frac{-1}{9\pi} - \frac{1}{9}$$

$$\underline{n=4} \quad a_4 = 0 + \frac{1}{16\pi} - \frac{1}{16} \Rightarrow a_4 = \frac{1}{16\pi} - \frac{1}{16}$$

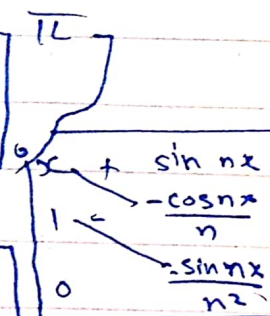
$$\underline{n=5} \quad a_5 = 0 + \frac{1}{25\pi} - \frac{1}{25} \Rightarrow a_5 = \frac{-1}{25\pi} - \frac{1}{25}$$

$$b_n = \frac{1}{\pi} \left[\int_{-\pi}^0 0 \sin(nx) dx + \int_0^{\pi} x \sin(nx) dx \right]$$

$$b_n = \frac{1}{\pi} \left[0 + \left[-\frac{x \cos(nx)}{n} + \frac{\sin(nx)}{n^2} \right] \right]_{0}^{\pi}$$

$$b_n = \frac{1}{\pi} \left[-\frac{\pi \cos(n\pi)}{n} + \frac{\sin(n\pi)}{n^2} - \frac{\sin(0)}{n^2} \right]$$

$$b_n = \frac{-1}{n} \cos(n\pi) + \frac{\sin(n\pi)}{\pi n^2} - 0$$



$$b_n = \frac{-1}{n} \cos(n\pi) + \frac{\sin(n\pi)}{\pi n^2}$$

$$n=1$$

$$b_1 = -\cos(\pi) + \frac{\sin(\pi)}{\pi} \Rightarrow b_1 = 0$$

$$b_2 = \frac{-1}{2} \cos(2\pi) + \frac{\sin(2\pi)}{4\pi} \Rightarrow b_2 = -\frac{1}{2}$$

$$b_3 = \frac{-1}{3} \cos(3\pi) + \frac{\sin(3\pi)}{9\pi} \Rightarrow b_3 = \frac{1}{3}$$

$$b_4 = \frac{-1}{4} \cos(4\pi) + \frac{\sin(4\pi)}{16\pi} \Rightarrow b_4 = -\frac{1}{4}$$

$$b_5 = \frac{-1}{5} \cos(5\pi) + \frac{\sin(5\pi)}{25\pi} \Rightarrow b_5 = \frac{1}{5}$$

$$f(x) = \frac{1}{2} \frac{\pi}{2}$$

09-1-24

Fourier sin series

$$f(x) = K \quad (0 < x < L)$$

$$\text{let } K = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi}{L} x\right)$$

$$b_n = \frac{2}{L} \int_0^L K \sin\left(\frac{n\pi}{L} x\right) dx$$

$$= \frac{2}{L} \left[K \left(\frac{-\cos\left(\frac{n\pi}{L} x\right)}{\frac{n\pi}{L}} \right) \right]_0^L$$

$$= \frac{-2K}{L} \left[\frac{L}{n\pi} \cos\left(\frac{n\pi}{L} x\right) \right]_0^L$$

$$= \frac{-2K}{n\pi} \left[\cos\left(\frac{n\pi}{L} x\right) \right]_0^L$$

$$= \frac{-2K}{n\pi} \left[\cos(n\pi) - \cos(0) \right]$$

C10-1
- marks que type
orthogonal exclude

15-1-24

imp

* non-homogeneous differential equation

$$\frac{dy}{dx} = \frac{2y - x + 5}{2x - y - 4} \quad \text{--- (1)}$$

$$\frac{a_1}{a_2} = \frac{b_1}{b_2}$$

$$x \rightarrow a_1$$

$$y \rightarrow b_1$$

$$x \rightarrow a_2$$

$$y \rightarrow b_2$$

$$\frac{a_1}{a_2} = \frac{b_1}{b_2}$$

$$\frac{-1}{2} \neq \frac{-2}{1}$$

let $x = x + h$

$$y = y + k$$

$$\frac{dy}{dx} = \frac{2(y+k) - (x+h) + 5}{2(x+h) - (y+k) - 4}$$

$$\frac{dy}{dx} = \frac{2y + 2k - x - h + 5}{2x + 2h - y - k - 4}$$

$$\frac{dy}{dx} = \frac{2y - x + 2k + 5 - h}{2x - y + 2h - k - 4} \quad \text{--- (1)}$$

$$2k + 5 - h = 0 \quad \text{--- (a)}$$

$$-k - 4 + 2h = 0 \quad \text{--- (b)}$$

multiply (b) by 2 and add in (a)

$$\begin{array}{r} 2x - h + 5 = 0 \\ -2x + 4h - 8 = 0 \\ \hline 3h - 3 = 0 \end{array}$$

$$3h = 3$$

$$h = 1$$

put $h=1$ in eq (a)

$$2x + 5 - 1 = 0$$

$$2x + 4 = 0$$

$$2x = -4$$

$$x = -2$$

put h & x in eq (1)

$$\frac{dy}{dx} = \frac{2y - x + 2(-2) + 5 - 1}{2x - y + 2(1) - (-2) - 4}$$

$$\frac{dy}{dx} = \frac{2y - x - 4 + 4}{2x - y + 2 + 2 - 4}$$

$$\frac{dy}{dx} = \frac{2y - x}{2x - y}$$

is now homogeneous eq

let $y = ux$

$$y' = u + x \frac{du}{dx}$$

$$u + x \frac{du}{dx} = \frac{2ux - x}{2x - ux}$$

$$u + x \frac{du}{dx} = \frac{x(2u-1)}{x(2-u)}$$

$$u + x \frac{du}{dx} = \frac{2u-1}{2-u}$$

$$x \frac{du}{dx} = \frac{2u-1}{2-u} - u$$

$$x \frac{du}{dx} = \frac{2u-1-2u+u^2}{2-u}$$

$$x \frac{du}{dx} = \frac{u^2-1}{2-u}$$

$$\frac{2-u}{u^2-1} du = \frac{dx}{x}$$

integrate

$$\int \frac{2}{u^2-1} du - \frac{1}{2} \int \frac{2u}{u^2-1} du = \int \frac{dx}{x}$$

$$2 \int \frac{1}{u^2-1} du - \frac{1}{2} \int \frac{2u}{u^2-1} du = \int \frac{dx}{x}$$

$$2 \tan^{-1}(u) - \frac{1}{2} \ln(u^2-1) = \ln x + \ln c$$

$$as \ y = ux$$

$$u = \frac{y}{x}$$

Bernoulli's Equation :

$$\frac{dv}{dx} + P(x) v = Q(x)$$

which is linear in v

Solⁿ $\frac{dy}{dx} + \frac{xy}{1-y^2} = xy^{-1/2}$

÷ eq by $y^{-1/2}$

RHS pay
y ki kol' power
hoga

$$\frac{dy}{dx} \times \frac{1}{y^{-1/2}} + \frac{xy}{1-y^2} \frac{1}{y^{-1/2}} = x$$

$$y^{1/2} \frac{dy}{dx} + \frac{x y^{1+1/2}}{1-y^2} = x$$

$$y^{1/2} \frac{dy}{dx} + \frac{xy^{3/2}}{1-y^2} = x$$

Step 2

$$\text{let } u = y^{1/2}$$

$$du = \frac{1}{2} y^{-1/2} dy$$

$$du = \frac{1}{2} y^{-1/2} dy$$

$$Q: \frac{dy}{dx} + \frac{xy}{1-x^2} = xy^{1/2} \quad \text{--- (A)}$$

Sol: divide ^(A) by $y^{1/2}$ on b.s

RHS par y ki
Koi power hogi

Step-1

$$\frac{1}{y^{1/2}} \frac{dy}{dx} + \frac{xy}{1-x^2} \frac{1}{y^{1/2}} = x$$

$$y^{-1/2} \frac{dy}{dx} + \frac{xy^{1/2}}{1-x^2} = x \quad \text{--- (B)}$$

let $u = y^{1/2}$

$$\frac{du}{dx} = \frac{1}{2} y^{-1/2} \frac{dy}{dx}$$

$$2 \frac{du}{dx} = y^{-1/2} \frac{dy}{dx}$$

using (B)

$$2 \frac{du}{dx} + \frac{x}{1-x^2} u = x$$

divide (2)

$$\frac{du}{dx} + \frac{x}{2(1-x^2)} u = \frac{x}{2} \quad \text{--- (C)}$$

this is a linear differential eq of order 1

$$\begin{aligned} \text{IF} &= e^{\int P(x) dx} \\ &= e^{\int \frac{x}{2(1-x^2)} dx} \\ &= e^{\frac{1}{2} \cdot \frac{1}{2} \int \frac{2x}{1-x^2} dx} \\ &= e^{\frac{1}{4} \ln(1-x^2)} \end{aligned}$$

$$\begin{aligned} \text{IF} &= e^{\ln(1-x^2)^{-1/4}} \\ \text{I.F} &= (1-x^2)^{-1/4} \end{aligned}$$

2.10
Q1 - Q10
Q17, 18, 19, 20

multiply I.F with (C)

$$(1-x^2)^{-1/4} \frac{du}{dx} + \frac{1}{2} \frac{x}{1-x^2} (1-x^2)^{-1/4} u = \frac{1}{2} x (1-x^2)^{-1/4}$$

$$\frac{d}{dx} (1-x^2)^{-1/4} u = \frac{1}{2} x (1-x^2)^{-1/4}$$

$$\frac{d}{dx} (1-x^2)^{-1/4} u = \frac{1}{2} \frac{x}{(1-x^2)^{1/4}}$$

integrate

~~$$\int \frac{d}{dx} (1-x^2)^{-1/4} u = \frac{1}{2} \frac{1}{-2} \int \frac{-2x}{(1-x^2)^{1/4}}$$~~

~~$$(1-x^2)^{-1/4} u = \frac{1}{-4} \ln(1-x^2)$$~~

$$\int \frac{d}{dx} (1-x^2)^{-1/4} u = \frac{1}{-2} \int \frac{-2x}{(1-x^2)^{1/4}} dx$$

$$(1-x^2)^{-1/4} u = \frac{1}{2} \int \frac{du}{-2u^{1/4}}$$

$$(1-x^2)^{-1/4} y^{1/2} = \frac{1}{-4} \frac{u^{-1/4+1}}{-1/4+1} + C$$

$$(1-x^2)^{-1/4} y^{1/2} = \frac{1}{-4} \frac{4}{3} u^{3/4} + C$$

$$(1-x^2)^{-1/4} y^{1/2} = -\frac{1}{3} (y^{1/2})^{3/4} + C$$

let $u = 1-x^2$

$$\frac{du}{dx} = -2x$$

$$\frac{du}{-2} = x dx$$

replace

2.9

Undetermined Coeff

Q: $3y'' + 10y' + 3y = x + \sin x$